Research with AI Agents

What Works, What It Costs, and What an Institute Needs

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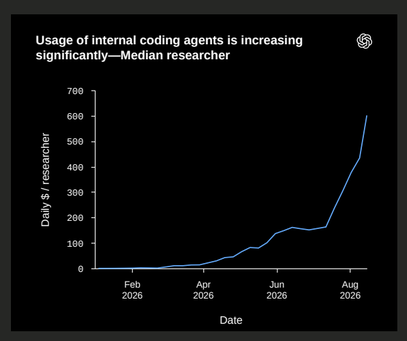
Outline

1. OpenAI and Agent Usage
2. Owning Intelligence and Infrastructure at Lamarr

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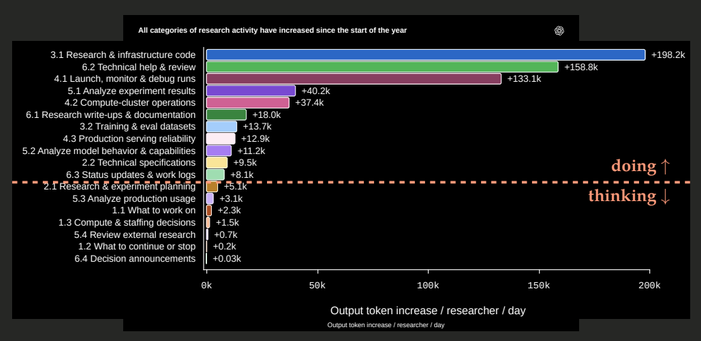
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FOMO Headline: OpenAI Agent Usage



Median researcher, daily $ spent on internal coding agents. Source: OpenAI, "Research acceleration" (self-reported).

What do agents do at OpenAI?



Increase in output tokens per researcher per day, by research activity. Source: OpenAI, "Research acceleration" (self-reported).

Dettmers' Interpretation

What the OpenAI chart suggests	Dettmers says
All 18 activities went up: agents are used everywhere, but how much and how exactly is still open.	Use agents or be left behind.
Agents took over the *doing*: code, runs, technical help.	"More than 90% of code and text should be written by agents."
Frequent, repetitive work grew most; rare decisions grew least.	Automate only if how often × time saved > time to build.
Did agents take over the *thinking*? We can't say: a plan or a decision is short, so tokens are the wrong measure.	Agent use is a *skill*: not everything can or should be automated (degree of automation).

Sources: timdettmers.com/2026/01/13/use-agents-or-be-left-behind; OpenAI, "Research acceleration" (self-reported).

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Field Report: Automating Exercises

1. **The problem.** PhD student, tutoring an exercise class every two weeks:
   - 2 h exercise class + 6 h preparation (solving the tasks, making the slides)
   - rough estimate: **2 h + 6 h = 8 h every two weeks**
2. **Automate as much as possible.**
   a) Task solving: an agent solves every task, with code and plots
   b) Slide creation: an agent turns the solutions into the class slides
3. **Left to do:** understand the solutions, make small updates to the slides.
   - class stays 2 h, preparation drops from 6 h to 2 h: **2 h + 2 h = 4 h every two weeks**
4. **Ask your agent to try it:** `github.com/Newida/AutomatingExercises` **Avoid:** Don't do anything BY HAND on this project. If you want to change it, ASK YOUR AGENT to change it.

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Automating at Bigger Scale

- **So far:** each of us automates our own tasks.
- **Share it:** one person builds it, everyone reuses and adapts it to their needs.
- **Institute level:** tasks that never pay off for one person start to pay off.

Example: travel expense reports

Collect receipts, fill in the university form, look up per-diem rules, fix what admin sends back: say **3 trips** × **3 h per person per year**. An agent that reads the receipts and files the form correctly needs ∼**200 h** (rules, forms, access to the admin system).

	how often × time saved	time to build	
one person	$3 \times 3\,\text{h} = 9\,\text{h/year}$	200 h	**never pays**
institute, 10 000 people	$10\,000 \times 9\,\text{h} = 90\,000\,\text{h/year}$	200 h	**pays >400×**

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What Agents Cost

- **Me:** Claude *and* Codex, 20 € plan each: **40 € per month**. One plan alone is not enough.
- **Colleagues:** 100 € Claude plan, and they hit its limit **every week**.
- **As we have seen:** every Lamarr member needs AI agents. You cannot work without them anymore.

Dear management: do not assume that employees will continue spending their own money. We are bottlenecking ourselves: we don't want to spend 100 € per month just to do our work, so we spend 20 €, effectively leaving performance on the table. Also, people will just switch to other institutes, even outside Germany, that supply them with the AI they want and need, for free, to get important work done.

So let's talk about costs.

Decision 1: Subscription or API?

Subscription (20–200 €/month)

- \+ flat price, no surprise bills
- \+ very cheap for heavy interactive use
- – usage caps (5-hour and weekly)
- – one person, one login; not for pipelines

API (pay per token)

- \+ no caps, run many agents in parallel
- \+ scripts, batch jobs, shared tools
- – heavy daily use gets expensive fast
- – needs a budget and someone watching it

Our comparison: one month of my usage, priced at API list prices

	subscription	same usage via API	factor
Claude	20 $ (Pro)	∼680 $	∼34×
Codex	20 $ (Plus)	∼145 $	∼7×

But one batch job, 27 benchmark tasks via API, cost only **2.25 $**.

Rule of thumb: people → subscription; pipelines and batch jobs → API.

Decision 2: Self-Hosting Open Models

- **Not for individuals**, but realistic for Lamarr: the institute already runs large GPU clusters.
- **Open-weight models:** Qwen (Alibaba), DeepSeek, GPT-OSS (OpenAI), Kimi (Moonshot), GLM (Zhipu), Mistral.
- **Go big, not small:** the frontier-class open models (0.5–1T parameters, e.g. DeepSeek, Kimi, GLM, Qwen-Coder) need ∼1 TB of GPU memory per copy: one or two full 8-GPU nodes, several of them to serve everyone.

Quality: our quantum-coding benchmark (27 tasks, solved), so far only small open models

Qwen, self-hosted on one GPU	10/27
GPT-OSS, self-hosted on one GPU	12–15/27
GPT-5-nano (small cloud model)	14/27
frontier cloud model (GPT-6.1-sol)	**26/27**

Small open models are clearly behind. The largest open models are much closer to the frontier; on our tasks they are **still untested**.

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Self-Hosting for the Institute: Pros and Cons

Quality and performance

- \+ data never leaves Lamarr (privacy, unpublished work, GDPR)
- \+ no vendor caps or outages, models don't change under you
- \+ fine-tune on our own data, research on the models themselves
- – even the best open models trail the closed frontier (by months), most on hard agentic tasks
- – slower when many people use it at once

Costs

- \+ no per-token bill: more use costs nothing extra
- \+ shared by all members, predictable budget
- \+ Lamarr already has the GPUs and the people who run them
- – a frontier-class model occupies whole nodes around the clock, taken from research jobs
- – power, cooling and staff to run it
- – hardware ages in 3–5 years; idle at night

Rough comparison

2 nodes × 300 k€ over 4 years ≈ 12.5 k€/month ≈ **300 people** × **40 €** ≈ **125 people** × **100 €**

With Lamarr's compute, a frontier-class open model for everyone is realistic: it pays off through privacy, scale and no caps; the closed models keep a lead of months.

Takeaways

For the individual

1. **You need to use AI agents.**
2. **Use AI agents to change your bottlenecks at work**
3. **Automate, to a degree, only what pays off:**
   how often × time saved > time to build
4. **Agent use is a skill** you need to learn.

For the institute

1. **Automate workflows for every individual:** what never pays off for one person pays off for 10 000.
2. **Provide your employees with agents.**
3. **Consider self-hosting.**